Forensic Fundamentals

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Introduction

Digital forensics is a key and specialist aspect of forensic science aimed at the systematic identification, retention, analysis and interpretation of information found within digital devices. It is described as the use of scientific knowledge and skills for recovering or analyzing materials found in electronic devices, typically for purposes of detecting and preventing computer crime (National Institute of Standards and Technology [NIST], 2020). Computing has made digital devices ubiquitous in all aspects of life and work. These devices vary from personal computers to mobile phones to removable storage media and networked complexity, all holding potentially invaluable evidence in criminal, civil and corporate investigations.

With the increasing importance of digital evidence, understanding the techniques and principles at the foundation of digital forensic practice is essential because what matters is reconstructing the activities that have occurred on a system and then accurately linking the events to specific users or processes. Unlike physical samples like DNA or fingerprints, which are processed in traditional forensic science, digital evidence is naturally fragile and unstable. When the system handles routine or automated processes, as well as when the users interact with the system or the background is on the system, important artefacts (timestamps, storage sectors, etc.) can be changed or erased, or overwritten entirely.

Due to the nature of evidence, this fragile aspect means that digital forensic investigations can be highly regulated by formal routines and statutory regimes which aim to ensure evidence retains its original integrity and is admissible (Casey, 2011). Failure to adhere to those principles risks contamination of the evidence or making the results untenable for the law. These processes involve the steps of obtaining evidence (often by forensic imaging), protecting it from tampering, analysing it with various systems, reporting the findings by means of an evaluative process and finally presenting the results in a thorough manner to a court or an organisation.

The project discussed here was a forensic analysis of a disk image acquired from a Windows operating system to find malicious or potentially dangerous activity against individual operating system users. The key findings included establishing the identity of the assigned user taking part in the observed activities, reassembling the kind and time of the system operation, and evaluating potential evidence being recovered through anti-forensic processes.

Since it was Windows-based, there is a major focus on its artefacts, which include user profiles and registry hives like NTUSER.DAT, Prefetch files, browsing history, file metadata, and deleted files restoration. For a well-rounded analysis, the open-source Autopsy was able to be used as a digital forensic tool for analysing disk images,

analysing Windows-specific artefacts, and reconstructing timelines due to its robust capabilities (Carrier & Spafford, 2021).

User Recognition and Attribution

The initial, arguably the crucial, activity in the investigation was to determine who accessed and used the system and to identify which account was related to the suspicious activities. In Windows environments, digital user profiles can be valuable forensic artefacts that enable exact user attribution (Microsoft, 2023).

Analysing the image of the disk, two user accounts, identified by the names Adam and Bob, were configured on the Windows machine. As the existence of multiple accounts made attribution difficult, and it became difficult to distinguish between artefacts created by users, forensic analysis soon turned up a striking imbalance of system activities in both profiles.

/img_FORFUNcw2021.E01/vol_vol2/Users — 4 Results

Table | Thumbnail | Summary

Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Location
Adam				2020-11-13 11:57:47 GMT	2020-11-13 11:57:47 GMT	2020-11-13 11:57:47 GMT	2017-08-30 12:29:57 BST	56	Allocated	Allocated	unknown	/img_FORFUNcw2021.E01/vol_vol2/Users/Adam
Bob				2020-11-13 11:47:49 GMT	2020-11-13 11:47:49 GMT	2020-11-13 11:47:49 GMT	2017-08-30 15:03:47 BST	56	Allocated	Allocated	unknown	/img_FORFUNcw2021.E01/vol_vol2/Users/Bob
[current folder]				2020-11-13 11:47:49 GMT	2020-11-13 11:47:49 GMT	2020-11-13 11:47:49 GMT	2009-07-14 04:20:08 BST	232	Allocated	Allocated	unknown	/img_FORFUNcw2021.E01/vol_vol2/Users/.
[parent folder]				2020-11-13 11:53:47 GMT	2020-11-13 11:53:47 GMT	2020-11-13 11:53:47 GMT	2020-11-13 11:44:51 GMT	56	Allocated	Allocated	unknown	/img_FORFUNcw2021.E01/vol_vol2/Users/..

Figure 1: Autopsy Users/Accounts list showing both Adam and Bob

Recent Documents

Table | Thumbnail | Summary

Source Name	S	C	O	Path	Date Accessed	Data Source
No preferred path found.lnk				No preferred path found	0000-00-00 00:00:00	FORFUNcw2021.E01
Desktop.lnk				C:\Users\Adam\Desktop	0000-00-00 00:00:00	FORFUNcw2021.E01
Downloads.lnk				C:\Users\Adam\Downloads	2017-08-30 13:36:09 BST	FORFUNcw2021.E01
Music.library-ms.lnk				C:\Users\Adam\AppData\Roaming\Microsoft\Windo...	0000-00-00 00:00:00	FORFUNcw2021.E01
Videos.library-ms.lnk				C:\Users\Adam\AppData\Roaming\Microsoft\Windo...	0000-00-00 00:00:00	FORFUNcw2021.E01
Pictures.library-ms.lnk				C:\Users\Adam\AppData\Roaming\Microsoft\Windo...	0000-00-00 00:00:00	FORFUNcw2021.E01
Documents.library-ms.lnk				C:\Users\Adam\AppData\Roaming\Microsoft\Windo...	0000-00-00 00:00:00	FORFUNcw2021.E01
Top secret.docx.LNK				C:\Users\Adam\Desktop\Top secret.docx	2018-09-25 09:04:08 BST	FORFUNcw2021.E01
images (2).lnk				C:\Users\Adam\Downloads\images.jpg	2017-08-30 13:36:32 BST	FORFUNcw2021.E01
Top secret.docx.lnk				C:\Users\Adam\Desktop\Top secret.docx	0000-00-00 00:00:00	FORFUNcw2021.E01
images (3).jpg.lnk				C:\Users\Adam\Downloads\images (3).jpg	2017-09-12 12:12:56 BST	FORFUNcw2021.E01
images (3).lnk				C:\Users\Adam\Downloads\images (2).jpg	2017-08-30 13:37:01 BST	FORFUNcw2021.E01
portsmouth.jpg.lnk				C:\Users\Adam\Downloads\portsmouth.jpg	2017-09-12 12:14:04 BST	FORFUNcw2021.E01
secret gift.pdf.lnk				C:\Users\Adam\Downloads\secret gift.pdf	2017-09-12 11:48:48 BST	FORFUNcw2021.E01
Pictures.lnk				C:\Users\Adam\AppData\Roaming\Microsoft\Windo...	2017-08-30 13:39:11 BST	FORFUNcw2021.E01
Documents.lnk				C:\Users\Adam\AppData\Roaming\Microsoft\Windo...	2017-08-30 14:59:52 BST	FORFUNcw2021.E01
images.lnk				C:\Users\Adam\Downloads\images.png	2017-08-30 13:36:09 BST	FORFUNcw2021.E01
Peppa-Pig.jpg.lnk				C:\Users\Adam\Downloads\Peppa-Pig.jpg	2017-09-12 11:50:18 BST	FORFUNcw2021.E01
images (1).lnk				C:\Users\Adam\Downloads\images (1).jpg	2017-08-30 13:36:55 BST	FORFUNcw2021.E01
bbc_news_logo.png.lnk				C:\Users\Adam\Downloads\bbc_news_logo.png	2017-09-12 12:12:47 BST	FORFUNcw2021.E01
HxDSetupEN.zip.lnk				C:\Users\Adam\Downloads\HxDSetupEN.zip	2017-09-12 11:57:41 BST	FORFUNcw2021.E01
My Documents.LNK				C:\Users\Adam\Documents	2017-08-30 13:51:46 BST	FORFUNcw2021.E01
filetodownload,100391,en.pdf.lnk				C:\Users\Adam\Downloads\filetodownload,100391,en...	2017-08-30 14:58:05 BST	FORFUNcw2021.E01
secret.docx.LNK				C:\Users\Adam\Documents\secret.docx	2018-09-25 09:04:32 BST	FORFUNcw2021.E01
to do list.docx.LNK				C:\Users\Adam\Documents\to do list.docx	2018-09-25 09:03:36 BST	FORFUNcw2021.E01

Figure 2: Autopsy Recent Activity shows most activity from Adam

Adam's database contained plenty of artefacts consistent with sustained use of the system, all of this deliberate and targeted use of the tool. Application and Software The AppData directories were home to a large amount of app data, configuration files and cached files used in the running of routine and programmed operating systems. The

document directory showed some files, password-protected documents, and most of the shortcut or LNK files showed a frequent presence with files and applications. Windows registry artefacts, in particular the Most Recently Used (MRU) lists in Adam's NTUSER.DAT hive, which recorded repeated interactions for documents, executable resources and system components that were indicative of constant usage.

In complete contrast, Bob's user profile was very sparse, with little or no meaningful content in the directories. There was notably no browsing history, execution traces of applications or other registries that are typically associated with usage via the system. This drastic asymmetry of user data was key to demonstrating a behavioural difference between the two accounts. This imbalance indicated that Adam was the active user of the system, while Bob's account was a dormant or sporadic user.

Windows operating systems inherently produce many kinds of user-specific artefacts, and those artefacts offer a high confidence level of accuracy, thus giving digital investigators access to attributes for user actions in Windows operating systems. The NTUSER.DAT file from any given user is their own registry hive, which contains all the software execution history, file access timestamps, MRU lists and configs (Microsoft, 2023). Prefetch files complement the registry, which provides information on the running of the applications, such as how many times the operating system has run an executable and the last time it was executed. Together, these artefacts add up to a behavioural signature that can be linked to certain users.

Autopsy's Recent Activity module further supported the attribution by associating accessed documents, executed programs and browsing artefacts principally with Adam's user profile (Carrier & Spafford, 2021). Digging deeper into Adam's NTUSER.DAT hive also indicated entries indicating habitual running of software, often opening the system files, changing the app config, and other applications' configuration. These results confirmed user activity as steady and permanent rather than ephemeral or unintended user behaviour.

A consistency was observed in timestamps between registry entries, file metadata and Prefetch traces, which confirmed that Adam was the undisputed and active user of the system.

One key discovery was that Adam had password-protected Word documents in his documents folder. These protected files had not only been saved in a repository, but were also actively accessed, requiring specific identification to unlock. One of these was a document that had to be recognized by colour, while another required locating historical information about Dr Nick Savage, an ex-Staff member of the University of Portsmouth.

Notably, similar files or their associated artefacts were not in Bob's profile, which therefore did, to some extent, exclude him from making either access to, or interaction with, these files.

In order to pretend as Adam did his research and to gain evidence around the process, archived versions of the University of Portsmouth employees' webpage were accessed via the Internet Archive Wayback Machine. This library is used to find historical snapshots of web resources that are no longer available in the public domain (Internet Archive, 2024).

Through this searching process, it was also possible to find staff listings based on Dr Nick Savage's historical phone number, the Head of Computing. This material was successfully employed to unlock the password-protected document, substantiating Adam's previous external research process.

This behaviour illustrated more than just active system usage, but rather intentional problem-solving efforts that went beyond just looking at local data, including interactive retrieval of outside data.

Making Adam the only primary active user on the system was critical to the accurate interpretation of other forensic artefacts. With accurate user attribution, results of deletion patterns, file corruption and tool-assisted wiping utilities can be confidently attributed to deliberate user behaviour and not automatic processes or actions performed by other accounts of systems. Without such attribution, any inference on motivation, intention or anti-forensic activity would have been speculative and indeterminate.

Web Engagement and Evidence Of Tracking

Intentional web interaction aside, intent towards sites was indicated by browsing artefacts found in Adam's profile. Key findings include recovering, for example, that a Coremetrics tracking cookie was recovered from unallocated disk space.

Coremetrics technology, commonly used for analytics of commercial sites and transactional websites—especially on retail and service websites—enables us to track user interactions with online platforms and records their patterns of visits and interactions (Acxiom, 2023). The presence of this sign meant that Adam's web activities spanned more than just academic or informational browsing; they could even be transactional or interactive.

The placement of this Coremetrics cookie in unallocated space confirmed that it had been deleted over a period. But later, this was followed up with an investigation that revealed the same cookie to be recovered in the Windows/Cookies/Low directory, meaning that the site had been reviewed after there had been a deletion activity.

Modern browsers are capable of automatically restoring such tracking cookies even as the user deletes them from their browsing path, illustrating the persistent nature of browsing artefacts and their forensic detection capabilities (Mozilla Foundation, 2023).

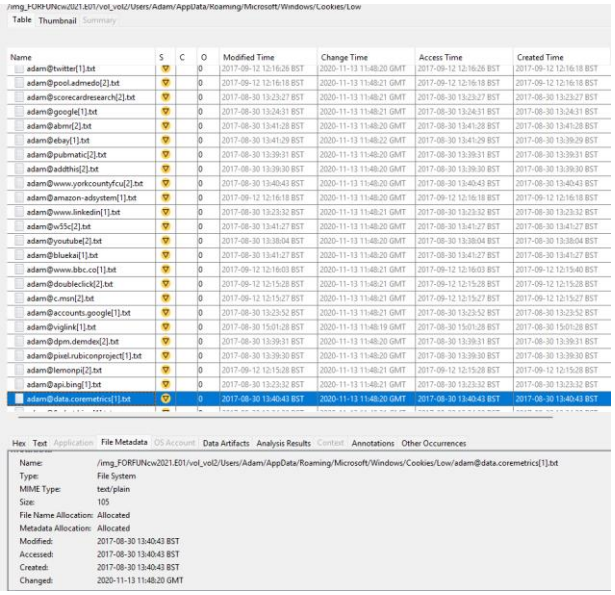


Figure 3: Cookies/Low Directory for restored Coremetrics File

Analysis of timestamps indicated that cookie creation and modification events correlated closely with contemporaneous document access and modification in Adam's profile. The connection between internet use and what files were viewed indicated that browsing activity was connected directly to tasks carried out inside the system, rather than casual or background internet use.

In fact, internet search behaviour correlated with attempts to use an encrypted file stored on the site.

Another example of proactive user-engagement was seen when Adam struggled to reach a University of Portsmouth page that was no longer accessible through regular means. Rather than throw it away, Adam opened an old version of the webpage via Wayback Machine and combed through the documents to locate staff who needed to be contacted to view password-protected documents. This browsing activity facilitated the freeing of secure records and demonstrated a conscious desire to obtain external information used in users' work.

Though some browsing artefacts had vanished, reviewing unallocated disk sectors uncovered remnants — deleted cookies, cached remnants, metadata — allowing a timeline of browsing activity to be reconstructed.

This case demonstrates the forensic maxim that attempts to delete or overwrite history and cookies are rarely exhaustive, with residual data in unallocated spaces providing vital investigative information.

Sensitive File Access Indicators

Several artefacts indicated Adam's access or manipulation of sensitive or protected data. The most important sign was the continuing operation of an app called UPDATER.EXE, noted in the Windows Prefetch directory.

Prefetch files, which are used to speed up application start times by placing data that needs to be fetched in order to facilitate application startup, sometimes inadvertently record forensic evidence such as the number of execute times, last run hours, and paths on the files (Microsoft, 2023). Therefore, Prefetch files offer investigators consistent measures of program usage patterns.

When UPDATER.EXE is discovered, the Prefetch files show that the executable has been repeatedly running. The system logs and artefacts generated with every run of this executable would appear along with the executable run, increasing the forensic capabilities in case of possible forensic discovery for this activity.

This observation became even more significant later upon the discovery of the deleted AgIFaultHistory.db file. Usually generated as a failure log in an application or system, this database file was discovered overwritten beyond recognition and had no legitimate organisation. Such overt corruption is rather rare and an unambiguous indication of a deliberate intent to tamper and destroy evidential value.

Since users rarely interact directly with fault history databases similar to this, the conclusion was that there was an intentional modification taking place.

Registry artefacts provided additional corroboration – MRU lists, Run keys and relevant application traces suggested repeated interactions with installers, executables, and sensitive documents inside Adam's profile.

The presence of password-protected Word documents and recurring execution of sensitive software applications support this reading.

If it took out the documents in secret, they also had to be discovered through independent investigation, and there was also evidence of other artefacts that were deleted or compromised, indicating tactical awareness of forensic persistence and an attempt to cover up traces of an act.

The Document Artefact: TopSecret.JPG

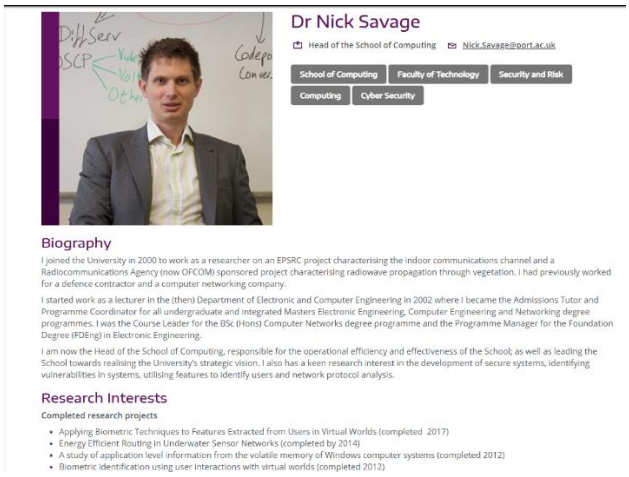
The presence of anti-forensic behaviour that was not detected was shown in a file named TopSecret.JPG. A closer look revealed this file as having been an extension to a regular JPEG image. In practice, but in-depth forensic analysis determined that true JPEG header signatures were lacking, which turned up the actual structure of documents for a Microsoft Word document (National Institute of Standards and Technology, 2018).

Such a mismatch between a file's extension and its internal format is a classic anti-forensic technique called file masquerading. File masquerading is designed to get around detection only by its surface appearances and postpone or suppress forensic examination.

How to use PowerPoint.pptx		0	2017-09-12 12:06:43 BST	2020-11-13 11:49:20 GMT	2017-09-12 12:06:43 BST	2017-08-30 13:08:17 BST	78096	Allocated	Allocated	unknown
secret.docx	!	0	2018-09-25 09:04:45 BST	2020-11-13 11:49:20 GMT	2018-09-25 09:04:45 BST	2017-08-30 13:51:46 BST	18432	Allocated	Allocated	unknown
Top secret.jpg	!	0	2017-09-12 11:46:42 BST	2020-11-13 11:49:19 GMT	2017-09-12 11:46:41 BST	2017-08-30 13:54:08 BST	18432	Allocated	Allocated	unknown
all my passwords.docx		0	2017-09-12 12:05:22 BST	2020-11-13 11:49:20 GMT	2017-09-12 12:05:22 BST	2017-09-12 11:58:51 BST	12153	Allocated	Allocated	unknown
UoP.docx		0	2017-09-08 10:45:17 BST	2020-11-13 11:49:20 GMT	2017-09-08 10:45:17 BST	2017-08-30 14:55:37 BST	11976	Allocated	Allocated	unknown

Users are less likely to find the sensitive document on a regular inspection or treat it as potentially irrelevant by hiding it as an innocuous image file.

TopSecret.JPG was password-protected and only accessible using system information already stored on the device. This was done through external information retrieval and was described above, primarily using Wayback Machine to access Dr Nick Savage's actual phone number from the University of Portsmouth webpages archived throughout his career.



Dr Nick Savage

Head of the School of Computing Nick.Savage@port.ac.uk

School of Computing | Faculty of Technology | Security and Risk | Computing | Cyber Security

Biography

I joined the University in 2000 to work as a researcher on an EPSRC project characterising the indoor communications channel and a Radiocommunications Agency (now OFCOM) sponsored project characterising radiowave propagation through vegetation. I had previously worked for a defence contractor and a computer networking company.

I started work as a lecturer in the (then) Department of Electronic and Computer Engineering in 2002 where I became the Admissions Tutor and Programme Coordinator for all undergraduate and integrated Masters Electronic Engineering, Computer Engineering and Networking degree programmes. I was the Course Leader for the BSc (Hons) Computer Networks degree programme and the Programme Manager for the Foundation Degree (FDEng) in Electronic Engineering.

I am now the Head of the School of Computing, responsible for the operational efficiency and effectiveness of the School; as well as leading the School towards realising the University's strategic vision. I also has a keen research interest in the development of secure systems, identifying vulnerabilities in systems, utilising features to identify users and network protocol analysis.

Research Interests

Completed research projects

- Applying Biometric Techniques to Features Extracted from Users in Virtual Worlds (completed 2017)
- Energy Efficient Routing in Underwater Sensor Networks (completed by 2014)
- A study of application level information from the volatile memory of Windows computer systems (completed 2012)
- Biometric identification using user interactions with virtual worlds (completed 2012)

Because this external data was crucial for unlocking the file, it corroborated the evidence of extensive multi-phase activity across local and foreign assets.

To disguise a publication by altering its extension has no legitimate organizational function and should be understood to be a tactical approach to hide publication.

But viewed in connection with deleted execution artefacts, corrupted logs, and subsequent system clean-ups, TopSecret.JPG looks like a part of a concerted anti-forensic effort, not an aberrant one.

Signs of Execution and Traces of Virtualisation

Further evidence suggested partial implementation and use of the virtual environment on the system as part of the investigation.

While no full virtual machine image was recovered, system directories contained the files associated with Microsoft Virtual PC, software that allows the creation of virtual machines in addition to host operating systems (Microsoft, 2023).

Virtual machines are frequently used to implement a technical protocol that isolates activity from the host operating system for possible testing and development or even nefarious reasons to obscure actions from forensic inspection.

Virtualisation artefacts indicated that the user had technical knowledge that went beyond that which was associated with ordinary computer use and that the compartmentalisation of activities was an intentional choice to block traceability.

This interpretation is reinforced by Prefetch analysis that identified execution traces for UPDATER.EXE but also for CCLEANER64.EXE, the 64-bit variant of CCleaner.

CCleaner is a popular utility for deleting browsing history, temporary files, cache, cookies, and registry keys (typically used to remove traces of user digital activity) (Piriform, 2024).

Interestingly, Prefetch files associated with these applications have been deleted and then regenerated upon closer inspection.

This pattern is quite highly suggestive that an effort to modify or distort execution records has occurred, as Prefetch files are recreated for every run of the underlying application.

$I30			0	2018-09-25 09:01:36 BST	2020-11-13 11:50:18 GMT	2018-09-25 09:01:36 BST	2017-08-30 12:25:03 BST	65536	Allocated	Allocated	unknown	/img_FORFUN
69.0.3497.100_CHROME_INSTALLE-B184BFBF.pf			0	2018-09-25 09:01:00 BST	2020-11-13 11:50:26 GMT	2018-09-25 09:01:00 BST	2018-09-25 09:01:00 BST	15954	Allocated	Allocated	unknown	/img_FORFUN
APPVSH~1.PF				2020-11-13 11:50:51 GMT	2020-11-13 11:50:51 GMT	2020-11-13 11:50:51 GMT	2020-11-13 11:50:51 GMT	0	Allocated	Allocated	unknown	/img_FORFUN
ATBROKER.EXE-5CD29207.pf			0	2017-11-17 10:18:19 GMT	2020-11-13 11:50:37 GMT	2017-11-17 10:18:19 GMT	2017-11-17 10:18:19 GMT	8672	Allocated	Allocated	unknown	/img_FORFUN
ATBROKER.EXE-5CD29207.pf.FileSlack			0	2020-11-13 11:50:37 GMT	2020-11-13 11:50:37 GMT	2020-11-13 11:50:37 GMT	2020-11-13 11:50:37 GMT	3616	Allocated	Allocated	unknown	/img_FORFUN
ATTRIB.EXE-58A07CAF.pf				2020-11-13 11:50:51 GMT	2020-11-13 11:50:51 GMT	2020-11-13 11:50:51 GMT	2020-11-13 11:50:51 GMT	0	Allocated	Allocated	unknown	/img_FORFUN
AUDIODG.EXE-AB22E9A6.pf			0	2017-10-02 12:43:07 BST	2020-11-13 11:50:41 GMT	2017-08-30 13:07:51 BST	2017-08-30 13:07:51 BST	25842	Allocated	Allocated	unknown	/img_FORFUN
AUDIODG.EXE-AB22E9A6.pf.FileSlack			0	2020-11-13 11:50:41 GMT	2020-11-13 11:50:42 GMT	2020-11-13 11:50:41 GMT	2020-11-13 11:50:41 GMT	2830	Allocated	Allocated	unknown	/img_FORFUN
AgAppLaunch.db			0	2017-08-30 12:25:06 BST	2020-11-13 11:50:31 GMT	2017-08-30 12:25:06 BST	2017-08-30 12:25:06 BST	334168	Allocated	Allocated	unknown	/img_FORFUN
AgAppLaunch.db.FileSlack			0	2020-11-13 11:50:32 GMT	2020-11-13 11:50:32 GMT	2020-11-13 11:50:32 GMT	2020-11-13 11:50:32 GMT	135168	Allocated	Allocated	unknown	/img_FORFUN
AgCx_S1_S-1-5-21-1748602848-2080315569-3795756			0	2017-09-12 12:16:38 BST	2020-11-13 11:50:34 GMT	2017-09-12 12:16:38 BST	2017-09-12 12:16:38 BST	2853159	Allocated	Allocated	unknown	/img_FORFUN
AgCx_S1_S-1-5-21-1748602848-2080315569-3795756			0	2020-11-13 11:50:34 GMT	2020-11-13 11:50:34 GMT	2020-11-13 11:50:34 GMT	2020-11-13 11:50:34 GMT	1753	Allocated	Allocated	unknown	/img_FORFUN
AgCx_S2_S-1-5-21-1748602848-2080315569-3795756			0	2017-09-12 12:18:00 BST	2020-11-13 11:50:20 GMT	2017-08-30 15:06:09 BST	2017-08-30 15:06:09 BST	3052349	Allocated	Allocated	unknown	/img_FORFUN
AgCx_S2_S-1-5-21-1748602848-2080315569-3795756			0	2020-11-13 11:50:21 GMT	2020-11-13 11:50:23 GMT	2020-11-13 11:50:21 GMT	2020-11-13 11:50:21 GMT	3267	Allocated	Allocated	unknown	/img_FORFUN
AgCx_SC3_0F505C221C17264E.db			0	2017-09-12 12:18:26 BST	2020-11-13 11:50:32 GMT	2017-09-12 12:18:26 BST	2017-09-12 12:18:26 BST	256771	Allocated	Allocated	unknown	/img_FORFUN
AgCx_SC3_0F505C221C17264E.db.FileSlack			0	2020-11-13 11:50:32 GMT	2020-11-13 11:50:33 GMT	2020-11-13 11:50:32 GMT	2020-11-13 11:50:32 GMT	1277	Allocated	Allocated	unknown	/img_FORFUN
AgCx_SC4.db			0	2017-11-17 10:19:33 GMT	2020-11-13 11:50:38 GMT	2017-08-30 15:29:39 BST	2017-08-30 15:29:39 BST	236574	Allocated	Allocated	unknown	/img_FORFUN
AgCx_SC4.db.FileSlack			0	2020-11-13 11:50:38 GMT	2020-11-13 11:50:39 GMT	2020-11-13 11:50:38 GMT	2020-11-13 11:50:38 GMT	994	Allocated	Allocated	unknown	/img_FORFUN
AgGlFaultHistory.db			0	[illegible]	[illegible]	[illegible]	2017-08-30 12:27:21 BST	391086	Allocated	Allocated	unknown	/img_FORFUN
AgGlFaultHistory.db				0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0	Unallocated		unknown	/img_FORFUN
AgGlFgAppHistory.db			0	2018-09-25 09:05:19 BST	2020-11-13 11:50:45 GMT	2017-08-30 12:27:21 BST	2017-08-30 12:27:21 BST	350379	Allocated	Allocated	unknown	/img_FORFUN
AgGlGlobalHistory.db			0	2018-09-25 09:05:19 BST	2020-11-13 11:50:49 GMT	2017-08-30 12:27:21 BST	2017-08-30 12:27:21 BST	828137	Allocated	Allocated	unknown	/img_FORFUN
AgRobust.db			0	2018-09-25 09:05:19 BST	2020-11-13 11:50:51 GMT	2017-08-30 12:27:20 BST	2017-08-30 12:27:20 BST	262040	Allocated	Allocated	unknown	/img_FORFUN
CCLEANER.EXE-DD6AC7E5.pf			0	2017-11-17 10:18:28 GMT	2020-11-13 11:50:36 GMT	2017-11-17 10:18:28 GMT	2017-11-17 10:18:28 GMT	24648	Allocated	Allocated	unknown	/img_FORFUN

There were 0 datasource(s) found with occurrences of the MD5 correlation value

Figure 4: Prefetch directory view containing multiple installation-related .pf entries, indicating attempted setup of virtualisation components or system-level installers.

The presence of multiple Prefetch entries deleted and later rebuilt indicates manipulation of forensic timelines.

Moreover, corrupted header structures of the disguised TopSecret.JPG file, in line with intentional overwrite, suggest organised manipulation of the system.

In conjunction with evidence that the stack has perhaps increased in both virtualisation and the use of sophisticated applications, these findings speak to increased technical awareness and intentional obfuscation.

Anti-Forensic Conduct and Obfuscation

During the forensic investigation, there was definite, clear evidence of Anti-Forensic tactics. These behaviours include intentional deletion, alteration, and manipulation of digital artefacts to undermine the integrity and recoverability of evidence.

The AgIFaultHistory.db file is one such example, not so much deleted, but rewritten to the point of making reconstruction of it impossible.

This overwriting significantly exceeds routine file deletion and represents active evidence destruction, an integral element of anti-forensic strategy.

Prefetch files related to UPDATER.EXE were recovered from non-allocated disk locations. While it wasn't the way Windows tends to delete those files on its own, cleaning could have been done manually or using specific cleaning tools.

The prefetch file, in particular, was found to be regenerated after deletion, showing that, even after deleting traces of deletion, the program was still running.

The process of deletion and regeneration illustrates efforts to strategically misrepresent execution history.

Execution of CCLEANER64.EXE was noted to occur soon after sensitive files were accessed.

CCleaner was already capable of removing traces of both systems and browsing, meaning that its usage indicated a deliberate strategy to remove forensic evidence.

However, the limitations of default deletion parameters in CCleaner hampered forensic recovery of residual data, further demonstrating that these anti-forensics measures are often an imperfect endeavour.

Strategy to Avoid Fraud

By examining the sequence and level of anti-forensic actions, I found that the approach employed was multi-layered and layered to enhance the effectiveness of concealment.

Early deletion selected high-value artefacts like fault data and Prefetch files.

Then, encrypted and obscured documents needed to be obtained from a third-party investigation to open, probably taking a bit longer to figure out what was there.

Lastly, CCleaner tried to reduce evidence that was left in the system, eg browsing history, temporary files, etc.

The fact that virtualisation artefacts also implied parallel or compartmentalisation operations, also meant to isolate sensitive behaviour.

This sequence is consistent with theoretical anti-forensic concepts, where a user uses multiple methods simultaneously to slow down the forensics or to make the attribution harder.

Manipulated image files, inconsistencies leading to steganography checks, corrupted document headers, and traces of virtualization all evidence deliberate, coherent, and technically competent concealment.

Conclusion

Ultimately, this investigation reconstructed an extensive and detailed account of system use and emerged with convincing evidence of deliberate anti-forensic behaviour.

Analysis of Windows artefacts indicated that Adam was an unequivocal primary user of the system.

The active handling of documents that were encrypted or disguised, his use of an external digital record to secure unlocking credentials, and repeating the same sensitive applications collectively speak to an intentionality of interacting with the secure material.

Efforts to delete, corrupt, and hide supporting artefacts of the offence—Prefetch files, fault history databases and document headers—are no doubt deliberate to interfere with forensic recovery.

The use of CCleaner and files associated with virtualisation demonstrates a mastery of processes that reveal a significant evidence concealment strategy.

In spite of these procedures, remnants of forensic traces existed which allowed reconstruction of timelines, user intention and application of multi-layered anti-forensic techniques.

Ironically, those same actions meant to clear away evidence proved to be corroborative evidence of conscious and informed engagement with forensics countermeasures.

Ultimately, the investigation showed that Adam had, technically, as well as strategically, the ability to manipulate the digital forensic narrative.

It highlights the complexities between two very different aspects of the multi-layered anti-forensic behaviours that must be considered in forensic examination and use of systems capable of addressing these challenges and maintaining the integrity of their investigation.

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